Reproducibility Report: Coppock (2017)

Did Shy Trump Supporters Bias the 2016 Polls? Evidence from a Nationally-representative List Experiment

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Archived at Zenodo: doi:10.5281/zenodo.24168301. This DOI always resolves to the latest release of the maintained version.

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This repository holds the actively maintained replication code for Coppock (2017), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1515/spp-2016-0005
Replication archive	10.60600/YU/Z88TZQ
Pre-analysis plan	None

The data are not redistributed here. The deposit is 3 files and 0.5 MB, and it lives in the Yale Dataverse rather than at Harvard, where most archives in this program sit. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. One file in `output/` is not committed. `trump.rds` is the deposited respondent-level file serialised, with nothing recoded, added or dropped, so committing it would redistribute the deposit by another route; `clean_trump.R` rebuilds it from `original/` in seconds. `ground_truth/` holds three things: the hand-classified list of every number the article prints, a script that runs the deposited archive itself and records what it produces, and a script that ties every published number to both that run and the rewrite. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_2017a.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 3 files, produces the published table, figure, appendix table and in-text quantities into `maintained/output/`, runs the deposited script 4 times, rebuilds the ground truth from all of it, prints every published number beside the sentence that states it, and verifies the deposit again. Required packages: tidyverse, estimatr, survey, list, rsample,

knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory.

The run that produced the committed output took 46 minutes, recorded step by step in `run_timings.csv`. Every duration quoted below is read from that file rather than estimated, so none of them can go stale; they are wall clock on one machine and indicative rather than a benchmark. 17.3 minutes of that is Figure 1, which refits 6 estimators in 25 subgroups across 2,000 bootstrap replicates and is almost the whole cost of the rewrite. Most of the rest is the deposited script, which does the same work more slowly and is run 4 times, at two replicate counts, under both of R's samplers, and against two sets of inputs. Anyone who wants only the rewrite can run the 7 scripts in `maintained/` and stop; the two `ground_truth/` scripts are the comparison against the deposit and against the article, and nothing in `maintained/` depends on them. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The CSV and TeX output comes back byte-identical; the PDF figure always shows as changed, because a PDF records the time it was written, so compare its PNG twin instead.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. The deposit is a single script, `shy_trump_analysis.r`, and it runs start to finish under R 4.6.2 with no change to any line of analysis. All 4 runs of it completed, with deprecation warnings and no errors; `ground_truth/archive_run_status.csv` records both of its passes, stripped to code and data and as shipped, as clean. One dependency is missing from its own commented `install.packages()` line, which names seven packages: the script also calls `arm::invlogit()`, namespace-qualified, so `arm` has to be installed and is never named. Every other package the script needs is on that line and every one of them installs and runs today, including `coefplot`, which supplies the `position_dodgev()` behind the figure.

Three things about the deposit matter more than whether it executes.

It writes nothing. There is not one uncommented call in the deposited script that saves a file: both tables are printed to the console as LaTeX and the figure is drawn to the active device, which under `Rscript` leaves a stray `Rplots.pdf` in the working directory and nothing else. Whatever a reader gets by running it has to be read off the screen. That decides what a checksum of the deposit can and cannot mean here, because there is no artifact of the archive for anyone to diff.

It ships with `sims <- 200` and a comment saying the article used 2,000, so what the deposit runs out of the box is not the configuration its own comment describes. The article never states a replicate count, which makes that comment the only record of one.

Its headline comparison is computed as `0.32547778 - tidy(fit_r)[2,2]`, with the direct-question estimate typed in as a literal rather than read from the data.

1.2 Does the maintained rewrite reproduce the paper?

Every point estimate, yes. Every bootstrap standard error, no, and neither does the deposit.

Of 392 recorded claims, 12 are numbers the article takes from outside the data (fivethirtyeight forecasts and election returns) and cannot be checked against the deposit, and 5 is a cell the article leaves blank. Of the 375 that remain, the rewrite matches 225 at the precision the article prints. Splitting that number is the whole story.

Point estimates and counts. 211 are checked. 35 of them are the income rows of Figure 1 and Table A1, where the deposited script attaches each block of numbers to the wrong quintile label; the rewrite prints the label the data supports, so it disagrees with the page by construction. See the Errata section. 3 more disagree, and each is the article disagreeing with itself or with its own data rather than with anyone's code: *Direct question, other or undecided (%)*, *Direct minus list experiment (pp)*, and *Points on the income scale entering the response model*. Every other point estimate, count and sample size in the article reproduces exactly.

Bootstrap standard errors. 124 are checked, leaving the income rows aside, and the rewrite matches 41. The deposited script does neither better nor worse: it matches 41 of the same 124, and the two draw the same bootstrap, cell for cell. The reason is given in the deposit's own header. The resampling code was rewritten in October 2019, two years after publication, when tidyverse bootstrapping moved to `rsample`, so the code that produced the published standard errors is not the code in the archive. No configuration of what is there recovers them, including the sampler R used before version 3.6.0. Every substantive conclusion in the article turns on whether an interval covers zero, and every one of those is unchanged.

Sentences with no number in them. 14 of the article's claims are about the shape, sign or count of its own estimates rather than about a value: that Whites were more likely to support Trump than any other category, that most of the unadjusted intervals contain zero, that the income groups did not diverge wildly. Nothing in a number-by-number comparison reaches them, and they are where a description can go on standing after the estimate beside it has moved. 10 of the 14 reduce to a verdict, and 10 of those hold. The remaining 4 name no threshold, so they get their evidence printed and no verdict.

One finding is neither a match nor a mismatch. The covariate-adjusted list experiment estimate is a nonlinear least squares fit, and it fails outright on 997 of the 2,000 bootstrap replicates. The deposited script and the rewrite handle those failures identically, by dropping them, so the adjusted intervals in the published Figure 1 and Table A1 rest on roughly half

the replicates the deposit's own comment names. Nothing in the comparison table shows this, because both sides agree; it is visible only by counting.

2 Paper overview

Coppock (2017) asks whether the 2016 polling miss can be explained by Trump supporters concealing their vote preference from pollsters, the "Shy Trump Supporter" hypothesis. A Reuters/IPSOS survey of 5,290 American adults, fielded 2 to 13 September 2016, asked vote preference directly and, later in the same instrument, ran a list experiment carrying the Trump item. If some supporters hide from the direct question, the list experiment should recover more support than the direct question does.

It recovers less. The direct question puts Trump support at 32.5%; the list experiment puts it at 29.6%. The article then looks for subgroups in which the two modes diverge, across party identification, education, income, gender, race and modeled vote propensity, unadjusted and covariate adjusted. Of 25 subgroups, 1 adjusted difference excludes zero. The article reports a combined estimator from Aronow, Coppock, Crawford and Green (2015) and the joint test of its assumptions, which the data fail.

The published quantities are Table 2 (the distribution of list responses), Figure 1 (the subgroup comparison), Appendix Table A1 (the same estimates as a table), and 29 statements of a quantity in the running text, several of which restate the same number in a different section. Table 1 is the wording of the two lists and carries no numbers, so it has no ground-truth row. Figure 1 plots exactly the estimates Table A1 prints, so the two share their rows.

3 Original archive reproducibility

The deposit is 3 files: a README, the respondent-level data as CSV, and one analysis script.

File	Bytes	Served as
README.txt	519	README.txt
shy_trump_analysis.r	11,439	shy_trump_analysis.r
shy_trump_cleaned.csv	481,797	shy_trump_cleaned.tab

All 3 checksums agree three ways: the MD5 Dataverse publishes, the MD5 of the bytes `?format=original` returns, and the MD5 of the copy this rewrite was developed against. `shy_trump_cleaned.csv` is ingested, so Dataverse serves it as `shy_trump_cleaned.tab` unless `?format=original` is passed; the published checksum describes the deposited CSV, not the derived tabular file, which is what makes the three-way agreement possible.

The deposit is not on Harvard Dataverse. It began in the Yale ISPS Data Archive under the handle `hdl:10079/zw3r2f9` and now lives in the Yale Dataverse under `doi:10.60600/YU/Z88TZQ`. The two locators are not interchangeable for a reader. The handle still resolves, but it resolves to `isps.yale.edu/research/data/d149`, which returns 404. The DOI is the only working pointer to the deposit, and it is the one recorded in `original_manifest.csv` and `download_original.R`. Yale Dataverse is a separate installation of the same software Harvard runs, so the file API and the `?format=original` convention are identical; only the host and the DOI prefix differ.

The archive runs. `shy_trump_analysis.r` executes to completion in a clean session, with its own analysis untouched. Its `setwd("")` line is commented out and marked "Uncomment to set working directory", so it reads its data from wherever it is run and has to be run from the folder holding the deposited CSV, or given that line. Four issues are worth recording.

Issue	Detail
Incomplete install line	The commented `install.packages()` names seven packages, which are exactly the seven the script loads with `library()`. It also calls `arm::invlogit()`, and `arm` appears nowhere in the line.
A dependency worth checking rather than assuming	The figure dodges with `position_dodgev()`, which `coefplot` exports and which originated in `ggstance`. `ggstance` is superseded by ggplot2's own support for horizontal geoms. `coefplot` is not: its current release is 1.2.9, dated 2025-08-23. Both install and both run, and the call is not broken.
Deprecations	`geom_errorbarh()` is deprecated in ggplot2 4.0; `gather()`/`spread()` and `do(data_frame(...))` are superseded. All still run.
Stray output	The script writes `Rplots.pdf` into the working directory, because the figure is printed to the default device rather than saved.

The deposit against its own README. The deposited `README.txt` describes exactly the two files that are there and says the script "conducts all analyses reported in the paper", which it does: Table 2, Figure 1, Table A1 and every number in the running text all come out of it, and the only published float it does not produce is Table 1, which is the wording of the two lists rather than an analysis. The one discrepancy is a letter. The README calls the script `shy_trump_analysis.R` and the deposited file is `shy_trump_analysis.r`. That

costs nothing on macOS or Windows and breaks a copied instruction on Linux, which is why every path in this repository is written to match the deposit exactly rather than to match its documentation.

The published number typed into the code. The deposited script computes the headline comparison as

```
0.32547778 - tidy(fit_r)[2,2]
```

The direct-question estimate is a literal. It agrees with the value the data give to within 7.7e-10, but nothing in the script checks that, and a reader cannot tell from the code where it came from. The rewrite computes both terms from the data.

The replicate count. The deposit ships

```
# The bootstrap sims take a while.  The article uses 2000 sims
# sims <- 2000
sims <- 200
```

so running the deposit as deposited does not run the configuration the comment describes. Both claims in that comment were tested rather than repeated, which is why the deposited script is run here at both replicate counts.

The first claim cannot be confirmed from the article. The article says its standard errors and intervals come from the nonparametric bootstrap and nowhere states how many replicates were drawn: not in the text, not in the note to Table A1, not in a footnote. The deposit's comment is the only statement of 2,000 anywhere in the record, and the rewrite adopts it for that reason rather than because the article corroborates it.

The second claim is a matter of measurement, and the answer is that neither count recovers the published standard errors and the two are hard to tell apart. Across the 120 standard errors Table A1 prints for the non-income subgroups, the deposited script at 2,000 replicates matches 39 and lands 0.13 points from the published figure on average; at the 200 it ships it matches 37 and lands 0.26 points away. The point estimates are unaffected either way, since they do not depend on the bootstrap. The shipped configuration is therefore not what makes the standard errors irreproducible, and "take a while" was 15.5 minutes at 2,000 replicates on the run that produced this report, against 1.2 at 200. The rewrite drew the same 2,000 replicates in 17.3 minutes.

How the archive is run here. `ground_truth/run_original_archive.R` copies the deposit into a scratch directory, because the script writes `Rplots.pdf` into its working directory and an archive that writes into itself can overwrite part of the deposit. It makes three edits to the copy and no others. Two of them fill in lines the deposit itself marks as configurable: the commented `setwd("")`, which the deposit says to uncomment, and `sims <- 200`, which the

deposit says the article set to 2,000. The third appends a block that saves objects the script has already computed. The `value_script` column of the ground truth is read out of that block, never transcribed.

Three of the four runs use a scratch copy holding the deposited script and the deposited CSV and nothing else. A script that runs against an archive as downloaded can pass by loading an intermediate the archive ships rather than one it built, and removing everything that is neither code nor data is the only way to tell the two apart. This deposit ships no intermediates, so the test passes trivially, but it is run and recorded rather than assumed. The fourth run is the same 200-replicate configuration against every file the deposit contains, and the two are asserted to return identical numbers, so the stripped runs are checked against their own control rather than left as a claim about intent.

Each run also records, by modification time, which deposited files it wrote over. That is the measurement behind the rule against running an archive in place, and here it comes back empty: the deposited script writes nothing at all, so running it in place would have damaged nothing. That is a fact about this deposit and not a general one, and it is cheap enough to measure on every run.

Run	Replicates	Sampler	Inputs	Status	Files left behind	Deposited files overwritten
current	2,000	Current	code and data only	completed	Rplots.pdf	none
rounding	2,000	Pre-3.6.0	code and data only	completed	Rplots.pdf	none
shipped	200	Current	code and data only	completed	Rplots.pdf	none
shipped_as_deposited	200	Current	as shipped	completed	Rplots.pdf	none

4 Errata

The deposited script mislabels the income quintiles, and the error reached the published Figure 1 and Table A1.

The script builds a factor whose levels are the values found in the data and whose labels are the names the article prints:

```
group = factor(group,
  levels = c(..., "income_quintile_20th - 40th Income Percentile",
                  "income_quintile_40th - 60th Income Percentile",
                  "income_quintile_60th - 80th Income Percentile",
                  "income_quintile_Above 80th Income Percentile",
                  "income_quintile_Below 20th Income Percentile", ...),
  labels = c(..., "Below 20th Income Percentile",
                  "20th - 40th Income Percentile",
                  "40th - 60th Income Percentile",
```

```
                    "60th - 80th Income Percentile",
                    "Above 80th Income Percentile", ...))
```

The levels are typed in alphabetical order, which is the order R would sort them into anyway. The labels are typed in substantive order. Every other block in the same call lines up, because for party identification, education, gender, race and vote propensity the alphabetical and substantive orders coincide. For income they do not, and the five income labels are rotated by one position. Each income row is printed under the label of the quintile below it, and the bottom quintile is printed as the top one.

The direction of the error is not a matter of interpretation. `income_quintile` is a self-describing character variable, and its five categories partition the deposited household income measure `USHHI2` into contiguous blocks in exactly the order their names claim:

income_quintile	USHHI2 range	N
Below 20th Income Percentile	1 to 4	804
20th - 40th Income Percentile	5 to 8	1038
40th - 60th Income Percentile	9 to 14	1361
60th - 80th Income Percentile	15 to 18	936
Above 80th Income Percentile	19 to 24	1151

`maintained/clean_trump.R` asserts this ordering and writes the table above, so the fact is checked on every run rather than asserted once in prose.

What changes. The published Table A1 row "Below 20th income percentile" reports N = 1,038 and a direct-question estimate of 30.7%. Those numbers belong to the 20th to 40th quintile. The bottom quintile has N = 804 and a direct-question estimate of 27.2%, which the article prints under "Above 80th income percentile". The same rotation appears on the income block of Figure 1.

What does not change. No estimate is wrong, no standard error is wrong, and no conclusion in the article turns on it. The article's claim about income is that "different income groups did not diverge wildly in their support," which is as true of the correctly labeled rows as of the rotated ones. What changes is which five-row block of numbers belongs to which quintile.

The maintained rewrite prints the labels the data supports. Those 64 ground-truth rows therefore carry `match_rewrite = FALSE` with a note recording that the disagreement is intentional.

5 What counts as a claim

Before anything is compared, the article and its appendix are read end to end and every numeric token is written down with its location, then classified by hand. That list

is `ground_truth/published_claims.csv`, 401 rows, and it is the only place in this repository where a published number is written down at all. Extracting exhaustively rather than searching for a pattern is the point: a search for bare digits misses a number inside a larger expression, and a search built around one notation misses everything written in another.

Class	Claims	Meaning
pipeline	366	The deposited data determines it, so the rewrite must produce it
descriptive	14	A claim about shape, sign or count rather than a value, needing a computed truth value
definitional	6	A scale endpoint, a category count, a null value being tested
structural	3	Field dates, an axis range: a fact about the study rather than a result
transcribed	12	Copied from a forecast, the election returns or another paper, and unable to drift

Every claim classified `pipeline` or `descriptive` is then checked twice, by two instruments that share no code. `ground_truth/build_ground_truth.R` joins it to the deposited script's output and to the rewrite's and records the verdict. `maintained/in_text_claims.R` carries the article's own sentence in a comment, recomputes the number from `maintained/output/` by its own path, and prints it beside the sentence in the order a reader meets it. Neither reads the other, and the build stops if a claim in the extraction is missing from either. Two paths to the same number are what makes a disagreement between them visible; one path can only be believed.

The remaining classes are exempt by construction rather than by omission. Nothing the pipeline does can move a field date, a scale endpoint, or a share of the vote in Wisconsin.

6 Number-by-number comparison

The ground truth carries one row per published number, with three values beside each other: what the article prints, what the deposited script produces when run today, and what the maintained rewrite produces. The second and third are read out of files rather than typed.

The published value is stored as the string the page prints, and a value counts as agreeing when printing it to that string's own precision gives that string's own digits. The comparison is between two pieces of text, not between a number and a tolerance. Two things follow. A published `3.0` is held to one decimal and a published `3` to none, a distinction a numeric published value cannot make, since `3.0` and `3` are the same double. A value sitting exactly on a rounding boundary is also decided by the same rule the typesetter used rather than by an inequality that happens to point the other way.

Block	Claims	Archive checked	Archive matches	Rewrite checked	Rewrite matches

Figure 1 and Table A1	325	325	227	325	180
Table 2	12	11	11	11	11
In-text	29	27	23	29	24
Descriptive	14	0	0	10	10
External to the data	12	0	0	0	0

Splitting the checked claims by whether they are point estimates or bootstrap standard errors separates the two kinds of disagreement.

Kind	Claims	Archive matches	Rewrite matches
Point estimates and counts	212	207	173
Bootstrap standard errors	153	54	42
Claims about shape, sign and count	10	0	10

Every row that does not match carries a `defect_locus` recording where the disagreement lives.

Defect locus	Rows	Meaning
archive	147	The deposited code, which either produced the published value and no longer exists in the deposit, or produced a label the data does not support
paper_internal	3	The article disagreeing with its own tables or with its own data

Every in-text quantity the article states is listed below with the value the rewrite produces.

Claim	Published	Rewrite	Match
Respondents surveyed (abstract)	5290	5290	yes
Direct question, Trump support (%), abstract	32.5	32.548	yes
List experiment estimate of Trump support (%), abstract	29.6	29.578	yes
Respondents surveyed (design)	5290	5290	yes
Direct question, Trump support (%)	32.5	32.548	yes
Direct question, Clinton support (%)	37.0	37.015	yes
Direct question, other or undecided (%)	30.5	30.438	no
Direct question, Trump support (%), restated in the design section	32.5	32.548	yes
List experiment, control group mean (items)	1.548	1.548	yes
List experiment, treatment group mean (items)	1.843	1.843	yes
List experiment estimate of Trump support (proportion)	0.296	0.296	yes
List experiment estimate of Trump support (%)	29.6	29.578	yes
Direct question, Trump support (%), restated in the results section	32.5	32.548	yes
Direct minus list experiment (pp)	2.9	2.97	no
Bootstrap SE of direct minus list (pp)	3.4	3.339	no
Share not reporting Trump support (%)	67.5	67.452	yes
Non-disclosers, control group mean (items)	1.610	1.61	yes
Non-disclosers, treatment group mean (items)	1.642	1.642	yes

List experiment estimate among non-disclosers (%)	3.3	3.272	yes
Subgroups compared	25	25	yes
Points on the party identification scale entering the response model	7	7	yes
Points on the income scale entering the response model	22	24	no
Combined estimate of Trump support (%)	34.8	34.755	yes
SE of the combined estimate (pp)	2.9	2.942	yes
Combined minus direct (pp)	2.2	2.207	yes
SE of combined minus direct (pp)	2.9	2.829	no
Joint test: list estimate among admitted supporters (%)	85.7	85.732	yes
Joint test: standard error (pp)	5.6	5.606	yes
Joint test: two-sided p-value	0.01	0.011	yes

Every claim the article makes about the shape of its own estimates follows, with what the estimates say.

Claim	Holds	Evidence
The list experiment estimate is below the direct question estimate	yes	Direct question 32.5 per cent against list experiment 29.6 per cent
The difference between the two modes is not distinguishable from zero	yes	Difference 3.0 points, 95 per cent bootstrap interval -3.7 to 9.2
The list estimate among non-disclosers is not distinguishable from zero	yes	Estimate 3.3 points, HC2 standard error 4.2 points, t = 0.77
Democrats are unlikely and Republicans likely to support Trump	yes	Direct question support: Strong democrat 4.5; Not very strong democrat 9.2; Lean democrat 13.0; Lean republican 73.4; Not very strong republican 68.1; Strong republican 90.4
More educated groups support Trump less	no threshold stated	Direct question support in published order: Less than high school 30.7; High school or some college 34.4; College 30.1; Graduate school 25.8
The least educated group is the smallest and its estimates the most uncertain	yes	Smallest subgroup is Less than high school at N = 157; its list experiment standard error is 18.6 points, the largest of any subgroup
Income groups did not diverge wildly in their support	no threshold stated	Direct question support: Below 20th income percentile 27.2; 20th-40th income percentile 30.7; 40th-60th income percentile 36.6; 60th-80th income percentile 34.0; Above 80th income percentile 33.9
Men and women did not diverge wildly in their support	no threshold stated	Direct question support: men 36.9, women 28.4
Whites were more likely to support Trump than any other racial or ethnic category	yes	White leads every other racial or ethnic category on 4 of 4 estimators
Likely voters expressed higher Trump support	yes	Likely voters above unlikely voters on 4 of 4 estimators
Most unadjusted difference intervals contain zero	yes	23 of 25 unadjusted difference intervals contain zero

The adjusted intervals shrink substantially	no threshold stated	Mean 95 per cent interval width, list experiment estimate: 30.5 points unadjusted against 21.5 points adjusted
The combined estimate is not distinguishable from the direct question estimate	yes	Combined minus direct 2.2 points, bootstrap standard error 2.8 points, t = 0.78
All but one adjusted difference interval contains zero	yes	1 of 25 adjusted difference intervals exclude zero: Hispanic

Three of the in-text numbers deserve a word, and all three are the article disagreeing with itself or with its own data rather than with anyone's code.

The article says the direct question and list experiment differ by "2.9 percentage points" while Table A1 prints 3.0 for the same quantity; 2.9 is the difference of the two rounded shares and 3.0 is the rounded difference, which is 2.97. The article likewise says 30.5% of respondents chose neither major-party candidate, which is 100 minus the two rounded shares; the weighted shares of the three remaining response categories sum to 30.4%.

The third is a label rather than an estimate. Equation 1 describes the income term as a "22-point Income" measure. The deposited `USHHI2` takes 24 distinct values, 1 to 24, with every value occupied, and it is the variable both the logistic and the NLS regressions use. Nothing downstream depends on the count, and no estimate moves; the article's description of its own covariate is two categories short.

7 Published Table 2

	Control list	Treatment list
0 items	0.11	0.11
1 item	0.37	0.22
2 items	0.40	0.46
3 items	0.13	0.15
4 items	0.00	0.06
N	2645	2645

Every cell matches the published table. The article leaves the control-list "4 items" cell blank; a three-item control list cannot produce four, so the rewrite prints 0.00.

8 Published Figure 1

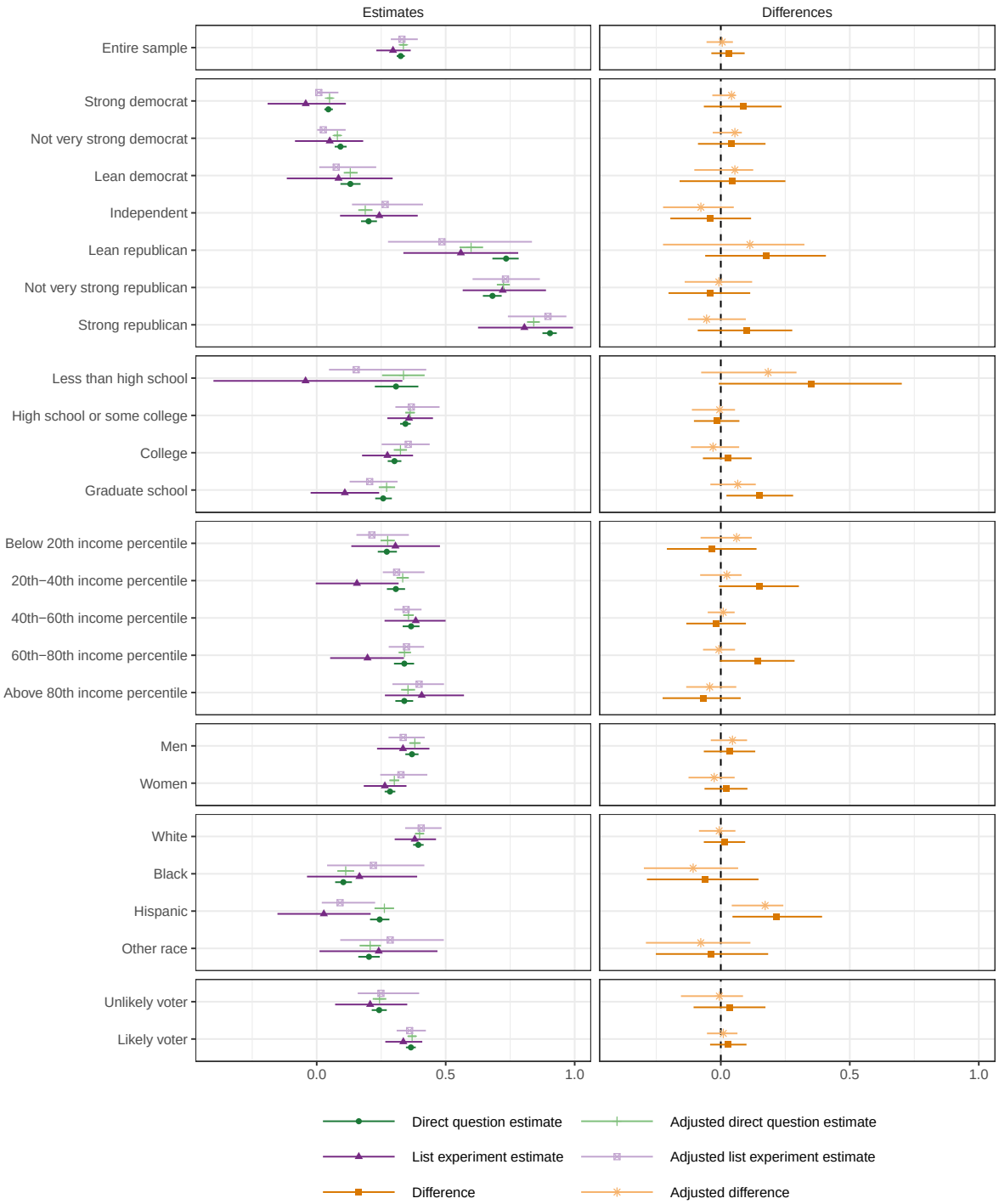


Figure 1: Direct and indirect estimates of Trump support, reproducing published Figure 1. Points are estimates, lines are 95 percent bootstrap percentile intervals from 2,000 replicates. The income block carries the corrected quintile labels described in the Errata section.

9 Bootstrap standard errors

The point estimates in Table A1 are deterministic given the data and reproduce exactly. The standard errors are not, and the reason is stated in the deposit's own header:

```
# Replication archive originally submitted to ISPS archive 2017-07
# Replication archive UPDATED 2019-10-13 because tidyverse bootstrapping moved to the "rsampl
```

The resampling code in the archive is not the resampling code that produced the published standard errors. The article appeared in June 2017; the deposited script was rewritten in October 2019 around `rsample`, and R's default sampler changed in R 3.6.0 in April 2019. Both samplers were therefore tested. The table below compares the published Table A1 against four runs: the deposited script at 2,000 replicates under the current sampler, the same under `RNGkind(sample.kind = "Rounding")`, the same at the 200 replicates the deposit ships, and the maintained rewrite. The five income subgroups are left out of the comparison, because the rewrite prints them under different labels and including them would charge the rewrite for the errata rather than for its arithmetic.

Run	Quantity	Cells	Match published	Mean abs. difference (pp)	Max abs.
Deposited script, 2,000 replicates	Point estimates	120	120	0.03	
Deposited script, 2,000 replicates, pre-3.6.0 sampler	Point estimates	120	120	0.03	
Deposited script, 200 replicates as shipped	Point estimates	120	120	0.03	
Maintained rewrite	Point estimates	120	120	0.03	
Deposited script, 2,000 replicates	Standard errors	120	39	0.13	
Deposited script, 2,000 replicates, pre-3.6.0 sampler	Standard errors	120	44	0.26	
Deposited script, 200 replicates as shipped	Standard errors	120	37	0.26	
Maintained rewrite	Standard errors	120	39	0.13	

Every point estimate reproduces under every configuration, which is what the first block of the table shows. The standard errors reproduce under none of them, and the pre-3.6.0 sampler is not the missing ingredient. It matches 44 cells to the current sampler's 39, which is marginally more, but it sits 0.26 points from the published table on average against 0.13, and its worst cell is 2.05 points out against 1.34. A few more exact hits alongside a wider tail is what a different random draw looks like, not what a recovered procedure looks like. The drift is therefore the change of resampling code that the deposit's own header records rather than the change of sampler.

The rewrite and the deposited script agree exactly. Across all 240 non-income cells of Table A1, point estimates and standard errors alike, the largest absolute difference between the two is 0 and 240 cells are bit-identical. Both call `rsample::bootstraps()` on the same data at the same seed and the same replicate count, so the rewrite is not an approximation of the deposited bootstrap but the same draw of it. What the deposit cannot reproduce, the rewrite cannot reproduce either, and for the same reason.

Because the published standard errors cannot be recovered exactly by any configuration of the deposited code, the ground truth records them, compares them, and marks the disagreements rather than pretending to a match. Every substantive conclusion in the article rests on whether an interval covers zero, and every such conclusion is unchanged.

10 Maintained rewrite

10.1 Architecture

The deposit is one script that loads, estimates, plots and prints. The rewrite is 7 scripts plus shared definitions: one per published output, then one that computes truth values for the article's claims about shape, then one that prints every published number beside the sentence that states it and writes nothing. Two more in `ground_truth/` build the comparison. Every model is fitted in the script that owns its output, and the scripts that check claims read and derive rather than refit, which is what keeps a second estimate of the same quantity from appearing anywhere.

Script	Produces
`maintained/helpers.R`	Packages, the subgroup label table, formatting helpers
`maintained/clean_trump.R`	`output/trump.rds` and the income-quintile check, after testing the file against what the article describes
`maintained/table_2_list_response_distributions.R`	Published Table 2
`maintained/figure_1_direct_vs_list_estimates.R`	Published Figure 1 and the estimates behind it
`maintained/table_a1_subgroup_estimates.R`	Published Table A1, from Figure 1's estimates
`maintained/text_main_estimates.R`	The in-text quantities
`maintained/text_descriptive_claims.R`	Computed counterparts for the article's claims about shape, sign and count
`maintained/in_text_claims.R`	Every published number beside the sentence that prints it, in reading order
`ground_truth/run_original_archive.R`	The deposited script's own output, at both replicate counts and under both samplers
`ground_truth/build_ground_truth.R`	The number-by-number comparison

`figure_1_direct_vs_list_estimates.R` writes its estimates to CSV, and both `table_a1_subgroup_estimates.R` and `text_main_estimates.R` read that file rather than repeating the bootstrap. The published figure and the published table are the same numbers, and computing them twice is how the two come to disagree.

`clean_trump.R` does not recode anything, because the deposited CSV is already an analysis file. What it does is check: 5,290 rows, a binary assignment, a three-item control list and a four-item treatment list, every subgroup the article reports present in the data and no subgroup present that the article does not report, and the income quintiles ordered as their names claim. The last of these is the assertion that turns the errata into a fact the pipeline enforces.

10.2 Deprecated patterns replaced

Deposited	Rewrite	Why
`arm::invlogit()`	`plogis()`	Base R; drops a dependency
`position_dodgev()`, from `coefplot`	`position_dodge(width = 0.6)`	ggplot2 4.x dodges along a discrete y axis natively, which drops a dependency without changing the figure
`geom_errorbarh()`	`geom_linerange()`	Deprecated in ggplot2 4.0.0
`gather()` / `spread()`	`pivot_longer()` / `pivot_wider()`	Superseded
`do(data_frame(...))`	`nest_by()` + `reframe()`	`do()` and `data_frame()` are both retired
`xtable` + `print.xtable`	`knitr::kable` + `kableExtra`, plus CSV	Table output to a file rather than to the console
Figure printed to the device	`ggsave()` to PDF and PNG	The deposit saved no figure and left `Rplots.pdf` behind
`%>%`	`|>`	Native pipe
`rm(list = ls())`, commented `setwd()`	`here::here()`	Scripts run from anywhere
`0.32547778 - tidy(fit_r)[2,2]`	Both terms computed from the data	No published number is an input to a computation
`sims <- 200`	`sims <- 2000`	The replicate count the article used

One thing was deliberately left alone, and counting how often it fires turned up something worth a section of its own. The deposit wraps the list-experiment NLS in `try()` and lets a failed fit drop out of the average with `na.rm = TRUE`. The rewrite keeps both, because changing either would change the published estimator. What that guard is doing, though, is not incidental: see the next section.

`stargazer` and `commarobust` do not appear in this archive, and no meta-analysis is performed, so the substitutions those call for elsewhere in this program do not arise here.

10.3 How much of the bootstrap the adjusted estimates actually use

The covariate-adjusted list experiment estimate is the NLS regression of Imai (2011), fitted separately in every bootstrap replicate. It does not always fit. On 997 of the 2,000 replicates, 50 per cent of them, the fit fails outright, with a singular gradient or an exactly singular system, and that replicate contributes no adjusted estimate for any subgroup. The failure is all or nothing: 0 replicates lose some subgroups while keeping others, so every subgroup's adjusted quantities rest on the same 1,003 usable draws rather than on 2,000.

The reduced replicate count is a property of the published analysis, not of the rewrite. The deposited script has the same `try()` and the same `na.rm = TRUE`, and its adjusted standard errors are identical to the rewrite's cell for cell, so the intervals in the published Figure 1 and Table A1 for the two adjusted estimators and the adjusted difference rest on the same reduced set. Nothing here says those intervals are wrong. The finding is that the effective replicate count behind them is roughly half the nominal one, and that a reader comparing an adjusted interval to an unadjusted one is not comparing two bootstraps of the same size. `maintained/output/figure_1_bootstrap_diagnostics.csv` records the counts on every run, so the claim is checked rather than asserted.

11 Maintained rewrite verification

Quantity	Estimator	Cells	Match published	Errata rows
Point estimates	Adjusted difference	25	20	5
Point estimates	Adjusted direct question estimate	25	20	5
Point estimates	Adjusted list experiment estimate	25	20	5
Point estimates	Difference	25	20	5
Point estimates	Direct question estimate	25	20	5
Point estimates	List experiment estimate	25	20	5
Standard errors	Adjusted difference	25	3	5
Standard errors	Adjusted direct question estimate	25	16	4
Standard errors	Adjusted list experiment estimate	25	2	5
Standard errors	Difference	25	3	5
Standard errors	Direct question estimate	25	11	5
Standard errors	List experiment estimate	25	5	5

Each estimator is computed in 25 subgroups, of which 5 are the income rows that disagree with the published labels by construction.

The rewrite is deterministic. Every script that draws sets its seed first, nothing between the seed and the draw touches the random number stream, and running the pipeline twice produces byte-identical CSV and TeX output. The PDF figure differs between runs because a PDF records the time it was written; its PNG twin does not, and comes back byte-identical.

12 R environment

Package	Version
tidyverse	2.0.0
estimatr	2.0.1
survey	4.5
list	9.2.6
rsample	1.3.2
knitr	1.52
kableExtra	1.4.1
here	1.0.2

R version: R version 4.6.2 (2026-10-29). Platform: aarch64-apple-darwin23.